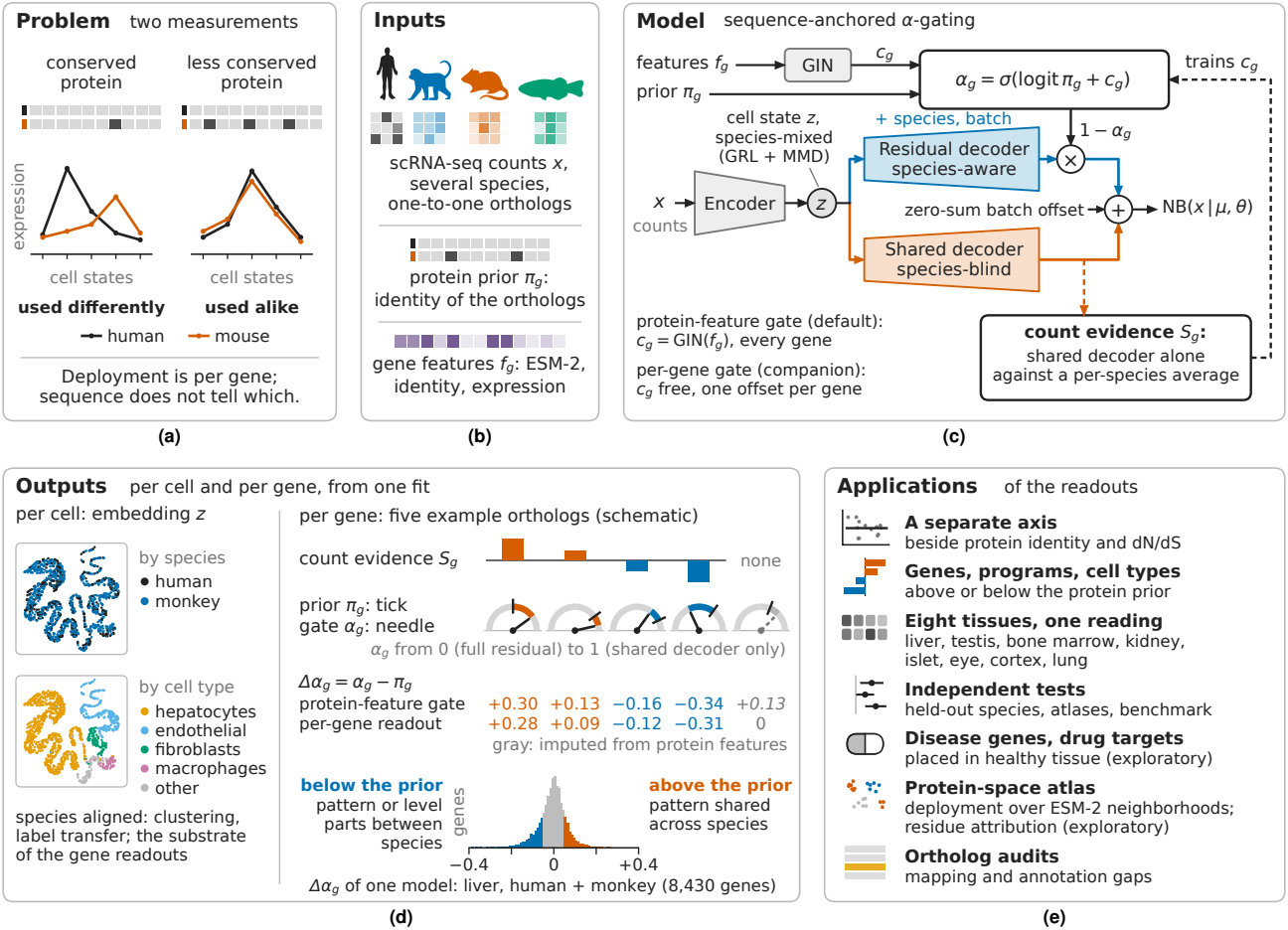

Problem two measurements
conserved protein
less conserved protein
expression
cell states
cell states
used differently
used alike
human
mouse
Deployment is per gene; sequence does not tell which.
(a)
Inputs
scRNA-seq counts x, several species, one-to-one orthologs
protein prior π_g: identity of the orthologs
gene features f_g: ESM-2, identity, expression
(b)
Model sequence-anchored α-gating
features f_g
prior π_g
GIN
c_g
$\alpha_g = \sigma(\text{logit}\,\pi_g + c_g)$
trains c_g
cell state z, species-mixed (GRL + MMD)
+ species, batch
$1 - \alpha_g$
Residual decoder species-aware
x counts
Encoder
z
zero-sum batch offset
$\text{NB}(x \mid \mu, \theta)$
Shared decoder species-blind
protein-feature gate (default): $c_g = \text{GIN}(f_g)$, every gene
per-gene gate (companion): c_g free, one offset per gene
count evidence S_g: shared decoder alone against a per-species average
(c)
Outputs per cell and per gene, from one fit
per cell: embedding z
by species
human
monkey
by cell type
hepatocytes
endothelial
fibroblasts
macrophages
other
species aligned: clustering, label transfer; the substrate of the gene readouts
per gene: five example orthologs (schematic)
count evidence S_g
none
prior π_g: tick
gate α_g: needle
α_g from 0 (full residual) to 1 (shared decoder only)
$\Delta\alpha_g = \alpha_g - \pi_g$
protein-feature gate +0.30 +0.13 −0.16 −0.34 +0.13
per-gene readout +0.28 +0.09 −0.12 −0.31 0
gray: imputed from protein features
below the prior
pattern or level parts between species
above the prior
pattern shared across species
genes
−0.4
0
+0.4
$\Delta\alpha_g$ of one model: liver, human + monkey (8,430 genes)
(d)
Applications of the readouts
A separate axis
beside protein identity and dN/dS
Genes, programs, cell types
above or below the protein prior
Eight tissues, one reading
liver, testis, bone marrow, kidney, islet, eye, cortex, lung
Independent tests
held-out species, atlases, benchmark
Disease genes, drug targets
placed in healthy tissue (exploratory)
Protein-space atlas
deployment over ESM-2 neighborhoods; residue attribution (exploratory)
Ortholog audits
mapping and annotation gaps
(e)